

PRANAV GANESH CHANDRATRE

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PROFILE

MSc Data Science candidate doing applied research on diffusion models, with a prior research Master's in physics. Sole contributor on a thesis that measures what a latent diffusion model's own sampling noise says about one specific image, and on the LoRA fine-tuning study that followed it. Took a multimodal audio-to-form system from hackathon prototype to a pipeline merged for clinical pilot, trained a PPO agent that beat its baseline $19.7\times$ in a 48-hour competition, and built tested data pipelines for model training and evaluation. Earlier physics work trained graph and dense networks on roughly 100,000 Geant4-simulated CMS HGCAL events across a multi-GPU Tesla P100 cluster. Interests: generative media, inference optimization, statistics of model evaluation.

Indian national on a UK Student visa, eligible for the UK Graduate route after the MSc. Available from September 2026 and open to industry, research, and internship opportunities, including positions requiring visa sponsorship.

EDUCATION

University of Birmingham — MSc Data Science | Sep 2025 – Sep 2026

Dissertation: Quantifying Epistemic Uncertainty in Diffusion Models for Chest Radiography, supervised by Dr Qingjie Meng, School of Computer Science.

IISER Kolkata — BS-MS Dual Degree, Physical Sciences | 2017 – 2022

Specialisation in Data Analysis and Machine Learning; GPA 8.52/10. MS thesis: Particle Detection and Classification in the High-Granularity Calorimeter of the CMS Detector, supervised by Dr Satyaki Bhattacharya (Saha Institute of Nuclear Physics) and Dr Ritesh Kumar Singh. INSPIRE Scholarship, Department of Science and Technology, Government of India.

SKILLS

Deep Learning & Generative: PyTorch, Hugging Face diffusers, latent diffusion, DDIM inversion, LoRA/PEFT, GANs, CNNs and U-Net, transformers, GNNs (GCN, GAT, residual), TensorFlow

Reinforcement Learning: PPO, Stable-Baselines3, Gymnasium, reward shaping, curriculum training

Statistics & Experiment Design: pre-registration, negative controls, bootstrap CIs, Friedman and Wilcoxon with BH correction, DeLong test, rank-biserial effect sizes, Negative Binomial GLM, rolling-origin CV, seed replication

Numerical Methods & Simulation: RK4, Verlet, Numerov, Euler-Maruyama SDEs, Monte Carlo sampling, QR and power iteration, finite differences, convergence analysis, Geant4

ML Engineering & MLOps: scikit-learn, XGBoost, LightGBM, PyTorch Geometric, GPU training, dbt, DuckDB, BigQuery, Docker, FastAPI, GitHub Actions CI, pytest, SQL

NLP & LLM Systems: Hugging Face, zero-shot classification (BART), LangChain multi-agent pipelines, structured extraction, speech-to-text, RAG, Gradio

Domains: medical imaging (DICOM, VinDr-CXR), experimental particle physics (Geant4, CMS HGCAL), quantitative finance and DeFi microstructure, forecasting, clinical NLP

RESEARCH & WORK EXPERIENCE

MSc Thesis — Quantifying Epistemic Uncertainty in Diffusion Models for Chest Radiography | Jun – Sep 2026

University of Birmingham | Sole contributor: architecture, experiments and write-up | Code: github.com/pranavgc/Uncertainty-Quantification-in-Diffusion-Models-for-Chest-Radiography

Stack: PyTorch, Hugging Face diffusers, latent diffusion (RoentGen-v2), DenseNet-121, statsmodels | Data: VinDr-CXR, 5 of 22 annotated findings, 512×512 | Compute: $2\times T4$, 16 GB each

- Built an inversion-anchored uncertainty estimator: DDIM inversion to a fixed latent ($\eta = 0$), then per-pixel variance over 20 stochastic reverse passes, so the estimate belongs to one patient's image rather than to the generator in general.
- Scored the maps against radiologist bounding boxes across 5 pathologies: 4 localise above chance, Consolidation highest at 0.694 pixel AUROC, while Cardiomegaly sits below chance at 0.450 because heart size is a global judgement rather than a local texture.
- Cut denoising steps from 30 to 15, halving the sampler passes behind a 39.6 s per-image generation and removing 32.4% of the variance for +0.016 AUROC; halving η removed 47% for +0.037. Most of the raw variance signal carries no spatial information.
- Designed an $N = 10$ negative control that varies only the estimator's sample count. It returned null on both measures, placing the effect in the sampling process rather than in the measurement.
- Benchmarked against MC-Dropout logit variance and Grad-CAM under dropout with Friedman tests, BH-corrected Wilcoxon and rank-biserial effect sizes. After a supervision confound in an earlier comparison was corrected, MC-Dropout still won 4 of 5 classes (Cardiomegaly $r = -1.00$), which puts the gap in label supervision rather than image exposure.
- Closed with a pre-registered 5-seed replication that withdrew one of the study's own significant findings: training-seed variability (SD 0.038) exceeded test-sampling variability (SD 0.028) and all 5 classes flipped sign. Reported as a result in its own right.
- Ran everything on two session-limited T4s with no persistent storage: pinned weights for byte-identical initialisation, contamination assertions that halt a run, and incremental artifacts so a timeout cost one run instead of ten.

LoRA Domain Adaptation of a Chest X-ray Diffusion Model — independent follow-up study | Sep 2026

Stack: LoRA/PEFT on the diffusion UNet, RoentGen-v2, Hugging Face diffusers, Sobel edge correlation | Code and technical report: github.com/pranavgc/Uncertainty-Quantification-in-Diffusion-Models-for-Chest-Radiography/tree/main/Extension

- LoRA fine-tuned the diffusion UNet (rank 8, 1.66M of 867M parameters trainable, 2,000 in-domain radiographs, captions stripped of class names so no label supervision could leak) to test whether weak pathology localisation was a domain-gap artifact.
- Edge dominance rose rather than fell: Spearman ρ against a Sobel edge map up 0.046 on Cardiomegaly and 0.061 on Consolidation across 150 prompt-matched pairs, both $p < 0.0001$, which ruled the domain gap out as the explanation.
- Pre-registered the test: a slope fitted before the intervention existed predicted Consolidation's AUROC shift to within 0.002 and got Cardiomegaly's sign wrong, so an observational relationship did not survive contact with an intervention.
- Peak variance still fell outside every annotated box, 0 of 150 images in both arms. Full fine-tuning is in progress.

AI Developer Intern — Armour Group (Clinical Form Extraction Pipeline) | May – Sep 2026

Stack: Python, microsoft/mai-transcribe-2 (speech-to-text), openai/gpt-6-astra-pro with Claude Opus 5 fallback | Grew out of the Patients Over Paperwork hackathon prototype (Armour Group Challenge, June 2026)

- Took the prototype through to implementation: the extraction pipeline was merged into the product codebase and submitted for a clinical pilot, built with clinicians and engineers who set the review thresholds and the failure cases.
- Added a second-pass reviewer for fields lost at segmentation boundaries and an evidence-verification stage for every extracted value, extending a six-stage multimodal pipeline that turns multilingual consultation audio into structured clinical forms (11+ form types, 38+ languages).
- Rebuilt the confidence engine around asymmetric clinical cost, cutting Brier score from 0.276 to 0.100 on the reference recording and raising high-confidence-tier precision to 0.93.
- Measured the veto layer rather than assuming it: it removes 60% of extracted fields and 72% of the errors, a 67% hit rate against a 48% base error rate.
- Grounded every field in a verbatim transcript span scored for presence, entailment and uniqueness, so an ambiguous span such as a bare "yes" is discounted.
- Built the batch evaluation harness and a threshold-fitting tool sweeping false-auto-fill rate against coverage, with transcription caching and deterministic sampling so evaluation runs repeat exactly.
- Split the model capability flag into separate temperature, structured-output and token-limit controls, which fixed reasoning-model calls that had been dropping JSON mode silently.

MS Thesis & Post-MS Research Intern — Particle Identification in the CMS High-Granularity Calorimeter | May 2021 – Jun 2024

IISER Kolkata & Saha Institute of Nuclear Physics | Stack: PyTorch, PyTorch Geometric, Geant4, scikit-learn, SciPy, NumPy, pandas | Compute: multi-GPU Tesla P100 cluster

- Trained dense, residual, graph-convolutional and graph-attention models on roughly 100,000 Geant4 events across 8 particle types, reaching 94.76% (ANN) and 96.66% (GNN) accuracy on shower-shape classification.
- Built the standalone Geant4 simulation behind the training data and the graph construction over calorimeter hits, wiring each fired cell to its six in-layer neighbours and the matching cell blocks in the layers either side, so the topology encodes detector geometry rather than a learned similarity.
- Found attention beat plain graph convolution on every feature set tested and residual connections helped both model families; adding simulated detector noise kept accuracy up under noisier inputs.
- Continued the work as a two-year post-MS research appointment and presented it as a poster at the IISER Kolkata Department of Physical Sciences foundation day, March 2023.

Generative Engine Optimisation Analyst — GradCore (client: Radiology Seminars) | Jun 2026

Stack: Python, Google Search Console, ChatGPT, Claude Sonnet 5, Gemini Pro, Microsoft 365 Copilot

- Showed that search rank does not predict whether an LLM names a provider: across matched prompts through four assistants, every model returned globally known institutions and dropped local providers holding page-one Google rank.
- Built a Python query-mining and gap-classification pipeline over 120 queries from 13 seed topics, labelling each against a four-competitor benchmark by fixed rules (71 flagged, 59.2%) and logging every prompt, output and source so the classification is auditable.
- Compiled and delivered the engagement's 83-page report and 27-slide presentation, working directly with the client throughout.

PROJECTS

AMM Liquidity Optimisation — Reinforcement Learning for Concentrated Liquidity Provision | Jun 2026

TurinTech AI "SaiFE Sandbox" challenge, UKFinnovator competition, Imperial College London — 48 hours, team of 3, second place in the challenge group | Stack: PPO, Stable-Baselines3, Gymnasium, NumPy | Code: github.com/pranavgc/Amm-Liquidity-Hunter-RL

- Trained a PPO agent for liquidity-range placement that returned +48.02% against +2.44% for the best baseline, a 19.7 $\times$ margin, profitable on 300 of 300 evaluation paths at Sharpe 3.87 with a worst path of +12.9%.
- Decomposed PnL across 300 trajectories before modelling anything: in-range fees ran roughly 26 against 4 of adverse selection, which reframed the problem from avoiding toxic flow to maximising range coverage.
- Caught the first agent (+24.4) matching the do-nothing baseline on all 12 reported metrics, then lifted it to +55.8 with dense in-range reward, an anti-HODL penalty and entropy at 0.05, and finished with a three-phase gas curriculum (0 $\rightarrow$ 1.5 $\rightarrow$ 3.0 over $\sim$ 5.4M timesteps). Evaluation ran at full gas with shaping disabled, so selection was never judged on the shaped signal.
- Overturned the team's own opening thesis in the behavioural analysis: the winning agent rebalances 291 times an episode and spends 87% of stake on gas, earning 5.30 per rebalance against 2.99 of gas cost, so the write-up was corrected rather than the data.
- Shipped the policy as a dependency-free NumPy forward pass to meet a 64 KB single-file limit: 4,931 parameters at 58.7 KiB, matching the PyTorch original to 1.02e-07.

Jaffle Demand Lab — Forecasting Pipeline with Ground-Truth Model Evaluation | Sep 2026

Stack: Python, dbt (Fusion), DuckDB, BigQuery, statsmodels, scikit-learn, LightGBM, pytest, GitHub Actions | Code: github.com/pranavgc/Jaffle-Demand-Lab

- Built a forecasting system where accuracy claims check against ground truth rather than another model: a seeded simulator (687k orders, 984k order lines, 1,096 days), a 23-model dbt layer with 114 tests, and a Python model suite that runs from a clean clone on DuckDB without cloud credentials.
- Cut RMSE 38.2% below baseline on a 90-day holdout scored once, benchmarking 9 models over $52,310 \text{ store} \times \text{SKU} \times \text{day}$ rows under rolling-origin cross-validation, and replacing a Poisson fit with a Negative Binomial GLM after measuring overdispersion (Pearson $\chi^2/\text{df} = 4.80$).
- Measured validation leakage instead of asserting it: random K-fold overstated accuracy by 6.1% against a size-matched rolling fold, with point-in-time correctness enforced by a dbt test that recomputes every rolling window.
- Injected five documented source faults (2,722 duplicate rows, a $100\times$ unit slip, negative-total refunds, MAR-missing weather, late-arriving rows), each paired with a detecting test and a repairing model, under CI on every commit.

Numerical Methods from Scratch — ODE, SDE, Eigenvalue and Boundary-Value Solvers | 2023, rebuilt Sep 2026

Stack: NumPy and Matplotlib only — no SciPy, no numpy.linalg, no library solvers | Code: github.com/pranavgc/Numerical_Physics

- Rebuilt 20 computational-physics problems with every algorithm hand-written: RK4 and Verlet, Numerov for the Schrödinger equation, Euler–Maruyama for SDEs including Black–Scholes, Box–Muller and inverse-transform Monte Carlo, modified Gram–Schmidt with QR iteration, power iteration with deflation, and shooting methods on a finite-difference Jacobian.
- Held each solver to its theory rather than to a plot: measured convergence orders of 2.000 (Verlet), 3.976 (RK4) and 0.498 (Euler–Maruyama) against theoretical 2, 4 and 0.5, with the Numerov ground state within $1.0\text{e-}11$ of the analytic infinite-well energy and the shooting solver recovering the exact initial slope to an rms of $9.1\text{e-}13$.
- Profiled and vectorised the slow paths: histogram binning from $O(N^2)$ to $O(N)$, and the random walk from 100 walkers in 8.9 s to 20,000 in 3.9 s, with the full suite running in about 59 s.

Predicting UK Restaurant Insolvency from Alternative Data | Jan – Mar 2026

Stack: Python, XGBoost (CUDA A100), scikit-learn, imbalanced-learn (SMOTE), Hugging Face Transformers, Google Places API, pandas | Code: github.com/pranavgc/Spill_the_tea

- Resolved 5.6M+ Companies House records into a 24,750-sample dataset by matching entities across four sources with no shared primary key: Companies House, the Food Hygiene Rating Scheme, ONS and Google Places.
- Added zero-shot sentiment classification (facebook/bart-large-mnli) for the consumer-facing signal and trained an XGBoost classifier (1,200 trees, CUDA A100) reaching $F1 = 0.24$, precision = 0.48 and AUC = 0.73 on the minority class under 95:5 imbalance.
- Showed via gain-based feature importance that structural and administrative signals beat customer-facing metrics as failure predictors, and that they are available before a restaurant opens.

SkillSync — AI-Powered Career Intelligence Platform | Apr – May 2026

Stack: Python, LangChain, Gemini Pro, Mistral-7B, FastAPI, Firestore, BigQuery, Docker, Cloud Run, React/TypeScript | Code: github.com/pranavgc/SkillSync

- Built a multi-agent LangChain pipeline that extracts a skill taxonomy from a CV, then researches the live job market and reports the gap, with Pydantic-constrained decoding holding LLM output to a schema so the deduplicated taxonomy is usable downstream.
- Served it through async FastAPI with Firestore and BigQuery behind it, containerised in multi-stage Docker builds on Cloud Run, with a React and TypeScript front end.

Other Projects

- Muon Scattering Tomography (Jun – Sep 2023): a U-Net segmentation CNN that localises civil structures and materials from atmospheric muon scattering in Geant4 simulation data, where the signal is what scattering angles imply about density along the path.
- CI/CD Workflows for MLOps (Dec 2024 – Jan 2025): ML pipelines triggering automatically on new data or code, plus a deployed Gradio chatbot, built with GitHub Actions, Hugging Face Spaces and GitHub Secrets.

CERTIFICATIONS

UKFinnovator Innovation Competition, UKfin+ (Imperial College London, June 2026) | Generative AI with Diffusion Models (NVIDIA Deep Learning Institute, October 2025) | LangChain Agentic AI and LangSmith Workshop (PwC, February 2026) | Sequence Models (DeepLearning.AI, 2024) | Generative Adversarial Networks Specialisation (DeepLearning.AI, 2023) | Convolutional Neural Networks (DeepLearning.AI, 2023) | Machine Learning Specialisation (Coursera / Stanford, 2023)

ADDITIONAL

Teaching Assistant, first-year physics laboratory — IISER Kolkata (Dec 2021 – Apr 2022). Volunteering: UoB Xtra Campus 3K and Brum Biosphere (2026).